

Demo: Homopolar Motor

Adult-controlled five-second AA-cell trials

Contact, force, reversal, and heat from timed evidence

Lesson Demo A · 20 minutes · after Lesson 4 — adult controls every contact

STOP LINE

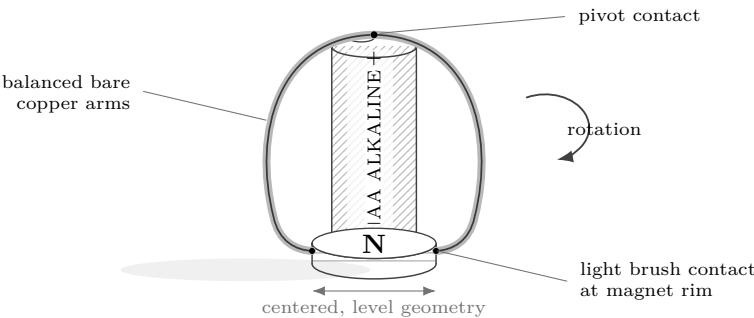
Use one AA alkaline cell only: never lithium, rechargeable cells, packs, or larger batteries. This apparatus is nearly a direct short. The adult makes and breaks contact, limits every trial to five seconds, and stops sooner for a stall, spark, odor, pain, wobble, damage, or unexpected motion. Cool between trials. Keep the magnet counted and away from small children, electronics, and anyone with an implanted medical device.

What you need

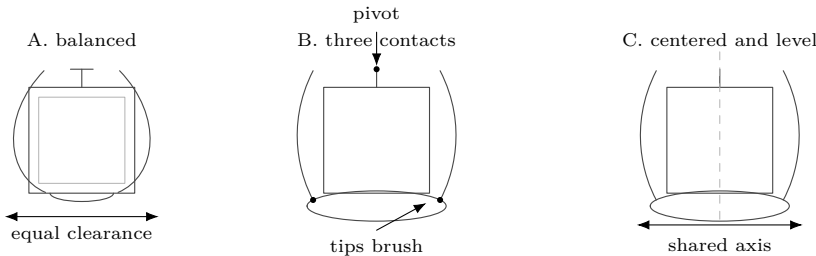
- One fresh AA alkaline cell and one room-temperature reference cell
- One unchipped neodymium disc magnet, 10–12 mm
- About 20 cm of bare copper wire, 18–20 AWG
- Pliers, timer, and paper direction flag

Predict first

Sketch the view from above. If the magnet flips, should rotation reverse? If both cell and magnet flip, should the result reverse or match the first trial? Where in the three-part circuit will friction and heating appear?



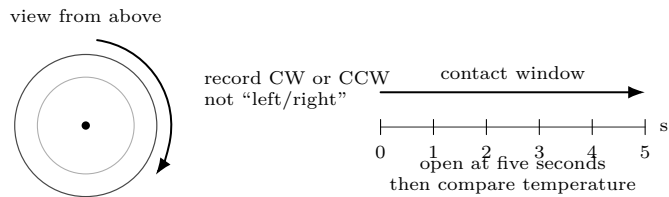
1. Shape and inspect — circuit open



1. Adult counts and inspects the single magnet, then mounts it on the flat negative end. Bend wire only while it is away from the cell.
2. Adult verifies symmetry, free clearance, a centered pivot, and two feather-light rim contacts. The learner points to the complete current path and identifies where excess pressure would add friction.

2. Measure one baseline

- 3. Put a paper direction flag behind the apparatus. Adult starts the timer and lowers the wire; learner calls “contact” at 0 s, direction as seen from above, start delay, and completed turns.
- 4. Adult lifts the wire at 5 s or at any stop condition. With the circuit open, adult compares wire, cell, and magnet with the reference cell using the back of a finger. Record *same*, *warmer*, or *stop*; never test temperature while current flows.
- 5. Cool at least 60 s. Stop for the day if any part is distinctly warm, uncomfortable, swollen, leaking, discolored, chipped, or damaged.



Trial	start delay	turns / 5 s	direction	wire	cell / magnet
Baseline	_____	_____	CW / CCW	same / warmer / stop	same / warmer / stop

3. Predict and test reversals

Keep geometry, time limit, stop line, and cool-down constant. Predict every row; adult tests only the minimum stable combinations needed to check the pattern.

Trial	Pivot terminal	Magnet face down	Prediction	Observed
A	+	N	baseline: _____	_____
B	+	S	reverse / same	_____
C	–	N	reverse / same	_____
D	–	S	reverse / same	_____

STABLE ASSEMBLIES ONLY

Reversing the cell requires remounting the magnet and reshaping the wire. If the magnet will not seat securely on the opposite terminal, skip C and D. Never add magnets, tape a live contact, hold a spinning wire, or lengthen a trial to obtain a result.

4. Troubleshoot with power open

Symptom	Inspect	One correction
No motion	Missing pivot or rim contact; coated wire	Expose clean copper; rebalance
Needs a push	Excess rim pressure or pivot friction	Open tips slightly; restore symmetry
Strong wobble	Unequal arms, off-axis pivot, tilted magnet	Recenter and level
Sparks or rapid heat	Heavy contact, damaged cell, short path	Stop; retire damaged parts; no retry
Direction changes	Intermittent contact or rocking magnet	Open circuit; rebuild geometry

Explain from the evidence

Current in the wire crosses the magnet's field and experiences a sideways force. The opposed arms turn that force into torque around the cell axis. Reversing either current or field reverses rotation; reversing both preserves it. Contact quality determines whether torque beats friction, not which direction the magnetic force points. Most input energy can become heat, which is why timing and open-circuit temperature comparisons are evidence, not optional housekeeping.

Final proof and closeout

Use the smallest safe set of trials to support: “one reversal reverses motion; two reversals restore it.” Name the compared trials, report direction from above and turns in five seconds, and state one result that would falsify the claim. Finish only after the adult confirms the circuit is open, every part is cool and undamaged, the magnet is counted into storage, and no wire touches the cell.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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